

XClarity Essentials Bootable Media Creator

Introduction to BoMC, how-to instructions, and text-mode examples

Lenovo

What is Bootable Media Creator?

Bootable Media Creator (BoMC) is a utility that can create bootable media to apply firmware updates offline.

Unlike previous versions, the bootable media creator does not support preboot diagnostic and operating system deployment function images for ThinkSystem servers, but this function is still supported for System x servers.

BoMC is available both as a GUI interface or a command-line interface.

Downloading BoMC

BoMC can be downloaded from Lenovo Support at <https://datacentersupport.lenovo.com/tw/zh/solutions/Invo-bomc>

Download Bootable Media Creator (BoMC) V12.4.0

Note: You must have administrator or root-equivalent operating-system privileges to use Bootable Media Creator.

Supported Lenovo Systems

- Lenovo ThinkSystem
- Lenovo ThinkServer
- Lenovo System x

Supported Lenovo Storage

- Lenovo Storage

Supported IBM Systems

- IBM Systems

XClarity Essentials Bootable Media Creator (BoMC) Version 12.4.0	
Invgy_utl_lxce_bomc01g-12.4.0_windows_i386	• Supported Operating Systems • Microsoft Windows 7/8/10

How to create bootable media with the BoMC GUI

Work through the following procedure to create a bootable media for a ThinkSystem server with the GUI-based BoMC utility.

Click each number in turn to see the procedure.

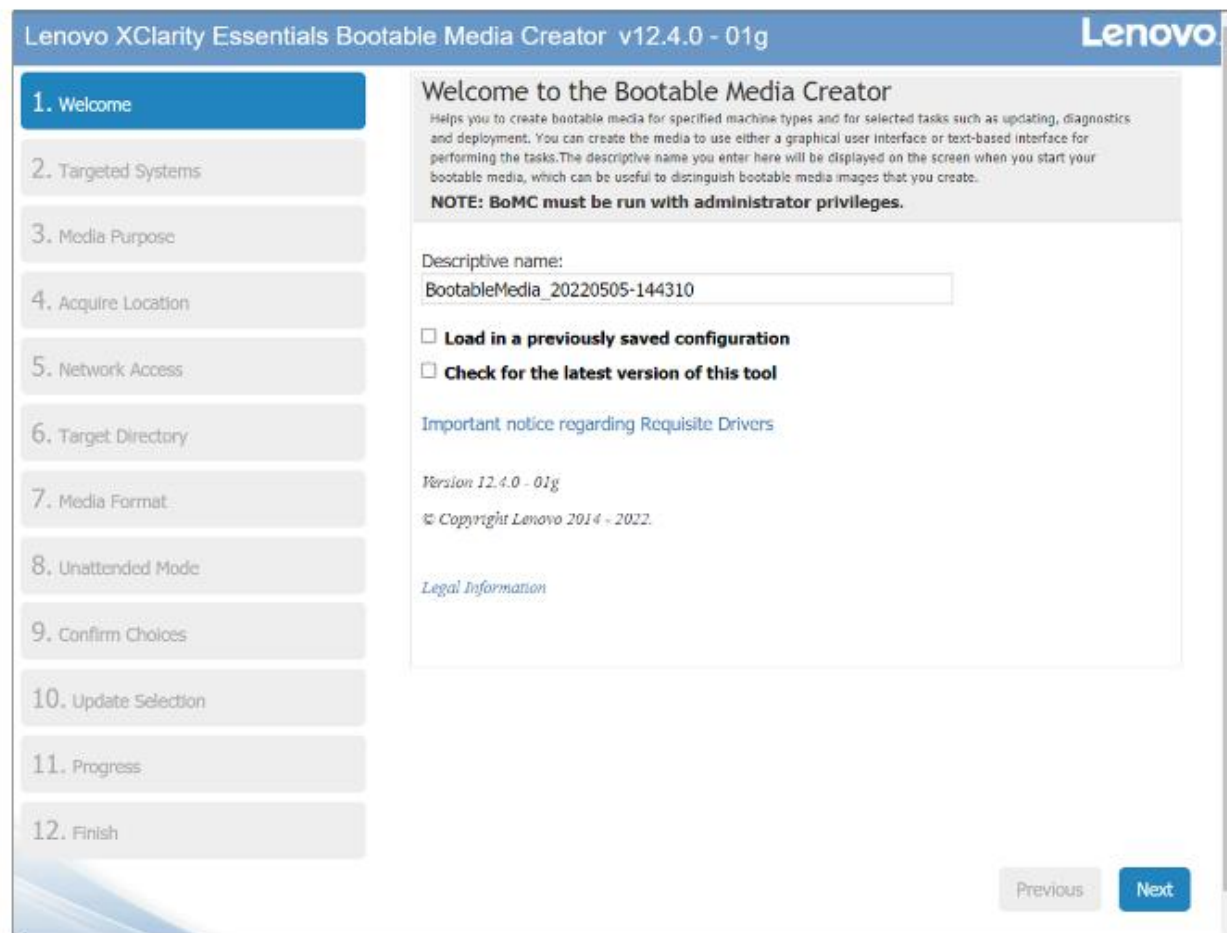


Step



How to create bootable media with the BoMC GUI

After downloading the LXCE BoMC utility from Lenovo Support, run it directly (no installation is required). When the welcome page is displayed, click **Next** to continue.



Step



How to create bootable media with the BoMC GUI

Select one or more ThinkSystem servers. A single bootable image can support multiple target machine types. The **Update List** button updates the system support list. The **Rollback** button rolls back to the original system support lists. After making a selection, click **Next** to continue.

1. Welcome
2. Targeted Systems
3. Media Purpose
4. Acquire Location
5. Network Access
6. Target Directory
7. Media Format
8. Unattended Mode
9. Confirm Choices
10. Update Selection
11. Progress
12. Finish

Targeted Systems

Specify what systems this bootable media should support. Please notice that you can only select systems in the same category.

Storage Platforms

Product Family	Machine Type
☐ D1212/D1224	4587
☐ D3284	6413

Lenovo Platforms

Product Family	Machine Type
☐ ThinkSystem SR850 V2	7D33
☐ ThinkSystem SR630 V2	7Z70
☐ ThinkSystem SR630 V2	7Z71
☒ ThinkSystem SR650 V2	7Z72
☐ ThinkSystem SR650 V2	7Z73
☐ ThinkSystem SR860 V2	7Z59

IBM Platforms

Product Family	Machine Type
☐ BladeCenter HX5	1909
☐ BladeCenter HX5	1910
☐ BladeCenter HS22	1911
☐ IBM BladeCenter HS23	1929
☐ BladeCenter HS22	1936

Update List **Roll Back**

Previous **Next**

Step



How to create bootable media with the BoMC GUI

Select the **Media Purpose**. You can create the BoMC package with firmware updates only, or the full package with firmware updates, VPD updates, inventory and FFDC collection, advanced system configuration, diagnostics, RAID configuration, and secure erase features.

Lenovo XClarity Essentials Bootable Media Creator v12.4.0 - 01g Lenovo

- 1. Welcome
- 2. Targeted Systems
- 3. Media Purpose**
- 4. Acquire Location
- 5. Network Access
- 6. Target Directory
- 7. Media Format

Media Purpose

Specify what you would like your bootable media to do. Based on what you select here, you will later be prompted for the appropriate information.

☒ **Update**
Will help you update the firmware on the machine types you choose. Note that this does not support IBM IntelliStation Pro, and it includes firmware updating only, and the drivers will need to be updated separately. The updating will be performed by LXCE UpdateXpress.

☐ **Enable Task AutoRun**
Allows a task to automatically begin when your bootable media is run on the target machine.

☐ **Full**
Will integrate all the functions in one image, including firmware update, VPD update, inventory and FFDC collection, advanced system configuration, diagnostics, RAID configuration, secure erase

☐ Look in a local directory for the RAID configuration file (*.ini) [viewSampleFile](#)

Step



How to create bootable media with the BoMC GUI

Follow the instructions to select the location from which you want to acquire files for the BoMC package.

Lenovo XClarity Essentials Bootable Media Creator v12.4.0 - 01g

Lenovo

1. Welcome

2. Targeted Systems

3. Media Purpose

4. Acquire Location

5. Network Access

6. Target Directory

7. Media Format

8. Confirm Choices

Acquire Location

You can choose to acquire updates, tools and bootable image from Lenovo support web site or from local directory.

☒ **Check the Lenovo support web site** - Downloads the appropriate updates automatically from the Lenovo support site.

☒ **UpdateXpress System Packs (UXSP's)** - UpdateXpress System Packs contain an integration-tested bundle of online, updateable firmware updates for each ThinkSystem / System X platform. This is the preferred method to obtain firmware for the server.

☐ **Latest available individual updates** - Check the Lenovo support web site for the latest individual version of each firmware package. This is the preferred method when you wish to install the latest updates or when Lenovo support instructs you to install the latest updates. This option may download newer updates and hotfixes, if available, than the UXSP option above.

☐ **Look in a local directory** - Specify a directory on the local file system containing specific individual files to include in the bootable media. The directory should have been populated with the required files either in a previous session of this tool, or manually. See [here](#) for specific requirements on manually obtaining required files.

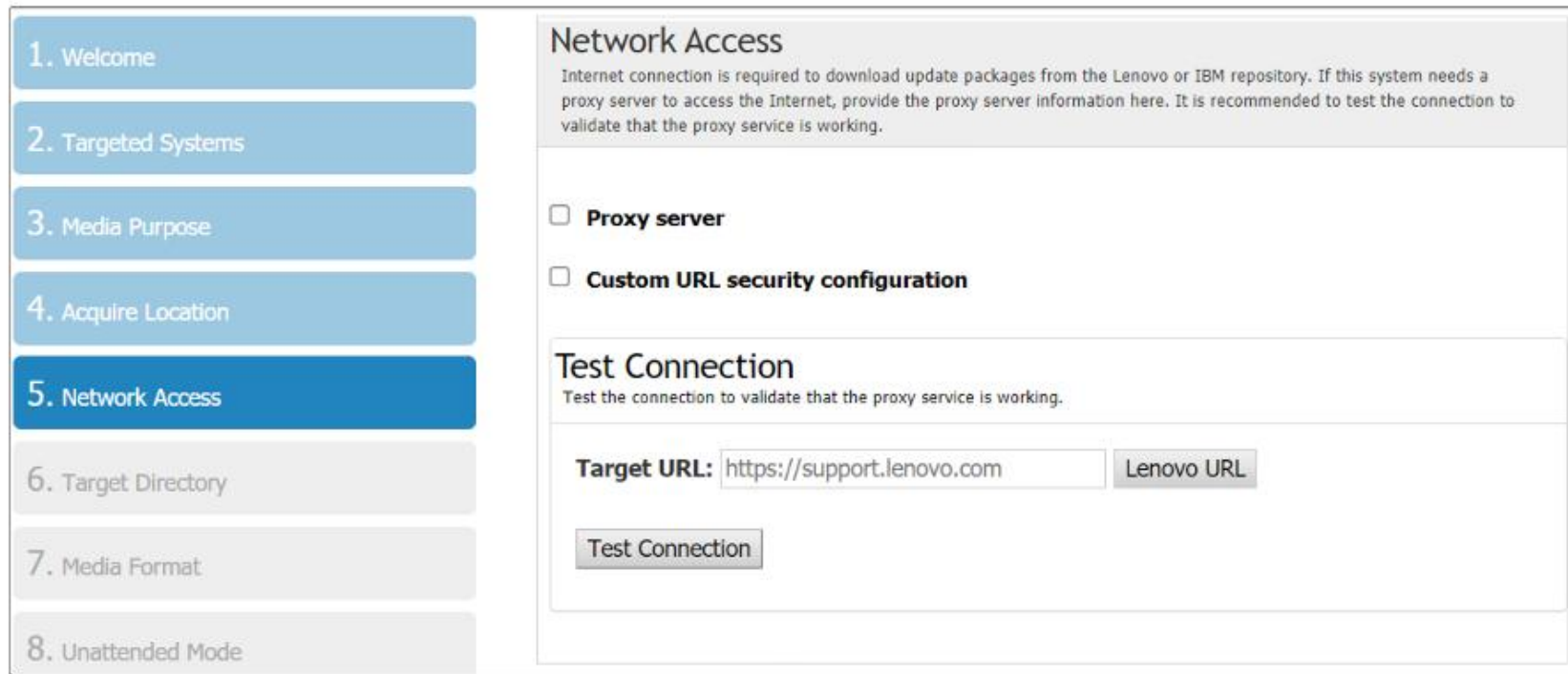
Browse

Step



How to create bootable media with the BoMC GUI

If necessary, set up a proxy server or custom URL security configuration to download update packages.



The screenshot shows the '5. Network Access' step in the BoMC GUI. On the left is a vertical sidebar with steps 1 through 8. Step 5 is highlighted in blue. The main area is titled 'Network Access' and contains a description, two unchecked checkboxes for 'Proxy server' and 'Custom URL security configuration', a 'Test Connection' section with a text input field containing 'https://support.lenovo.com', a 'Lenovo URL' button, and a 'Test Connection' button.

1. Welcome

2. Targeted Systems

3. Media Purpose

4. Acquire Location

5. Network Access

6. Target Directory

7. Media Format

8. Unattended Mode

Network Access

Internet connection is required to download update packages from the Lenovo or IBM repository. If this system needs a proxy server to access the Internet, provide the proxy server information here. It is recommended to test the connection to validate that the proxy service is working.

☐ Proxy server

☐ Custom URL security configuration

Test Connection

Test the connection to validate that the proxy service is working.

Target URL: Lenovo URL

Test Connection

Step



How to create bootable media with the BoMC GUI

Select the target directory for the updates and other resources files.

1. Welcome

2. Targeted Systems

3. Media Purpose

4. Acquire Location

5. Network Access

6. Target Directory

Target Directory

Indicate which directory on your computer you want to download the updates and other resource files to. The downloaded files are only needed during the media creation process. If you will not need these downloaded files later, then you can delete them after your media has been created, after this wizard is completed.

Directory:

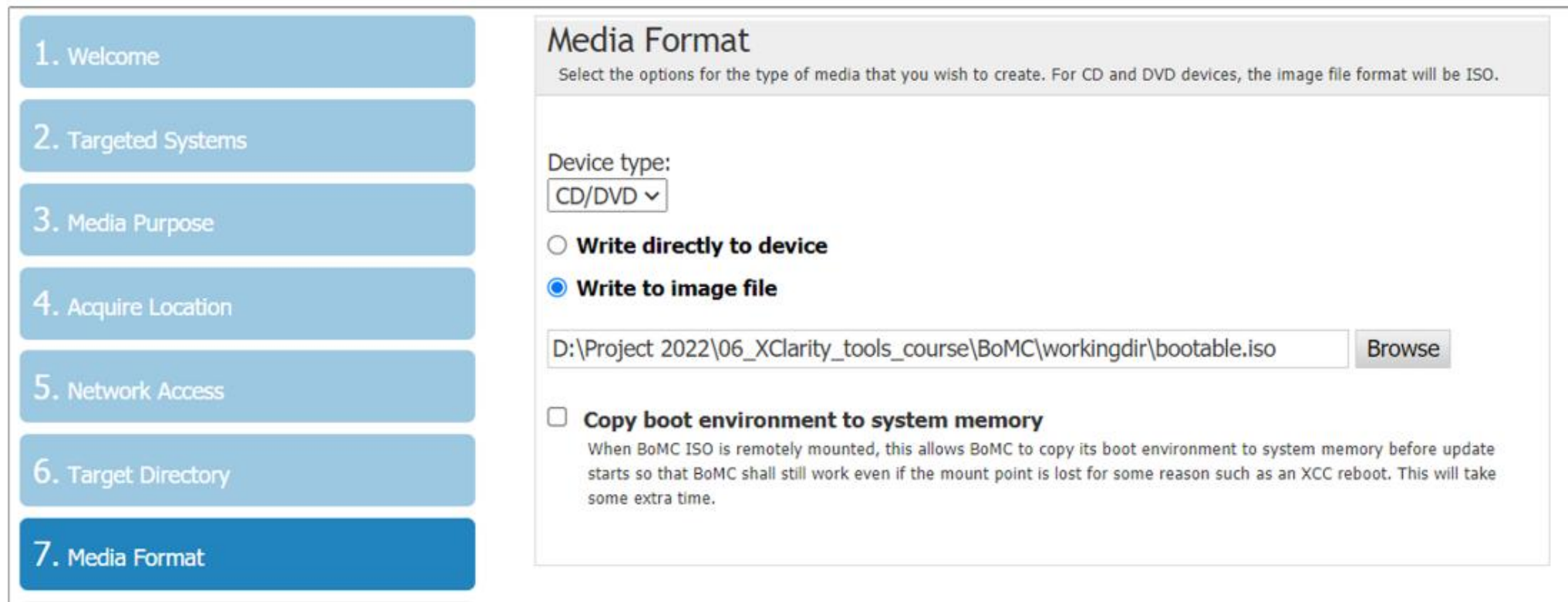
Browse

Step



How to create bootable media with the BoMC GUI

Select the format you want to use for the creation of the BoMC package.



The screenshot shows the 'Media Format' step in the BoMC GUI. On the left is a vertical sidebar with seven steps: 1. Welcome, 2. Targeted Systems, 3. Media Purpose, 4. Acquire Location, 5. Network Access, 6. Target Directory, and 7. Media Format (which is highlighted in dark blue). The main area is titled 'Media Format' and contains the following options:

- A subtitle: 'Select the options for the type of media that you wish to create. For CD and DVD devices, the image file format will be ISO.'
- 'Device type:' with a dropdown menu showing 'CD/DVD'.
- Two radio button options: 'Write directly to device' (unselected) and 'Write to image file' (selected).
- A text input field containing 'D:\Project 2022\06_XClarity_tools_course\BoMC\workingdir\bootable.iso' and a 'Browse' button.
- A checkbox option 'Copy boot environment to system memory' which is unchecked. Below it is a descriptive paragraph: 'When BoMC ISO is remotely mounted, this allows BoMC to copy its boot environment to system memory before update starts so that BoMC shall still work even if the mount point is lost for some reason such as an XCC reboot. This will take some extra time.'

Step



How to create bootable media with the BoMC GUI

If you selected **Update** on the **3. Media Purpose** screen, you can choose to use the unattended mode for the BoMC package. Refer to the description on the BoMC screen for more information about the unattended mode.

The screenshot displays the BoMC GUI with a vertical list of steps on the left and a configuration panel on the right. The steps are: 1. Welcome, 2. Targeted Systems, 3. Media Purpose, 4. Acquire Location, 5. Network Access, 6. Target Directory, 7. Media Format, and 8. Unattended Mode. Step 8 is currently selected and highlighted in a darker blue. The 'Unattended Mode Configuration' panel on the right contains a descriptive paragraph and three radio button options. The first option, 'Reboot BMC after updating BMC firmware automatically.', is checked with a blue square. The second option, 'Do not use unattended mode', is selected with a blue circle. The third option, 'Use unattended mode', is unselected with a grey circle.

1. Welcome

2. Targeted Systems

3. Media Purpose

4. Acquire Location

5. Network Access

6. Target Directory

7. Media Format

8. Unattended Mode

Unattended Mode Configuration

Helps you to create your created image able to run in a completely unattend mode. In that case, it will upload the log files onto your TFTP server, FTP server, your network file share(NFS/Samba) or your USB drive and shutdown your clients after the firmware update process. If you want to upload the log files onto the TFTP, FTP server, NFS or Samba server, please make sure the directory has been created and anonymous access granted.

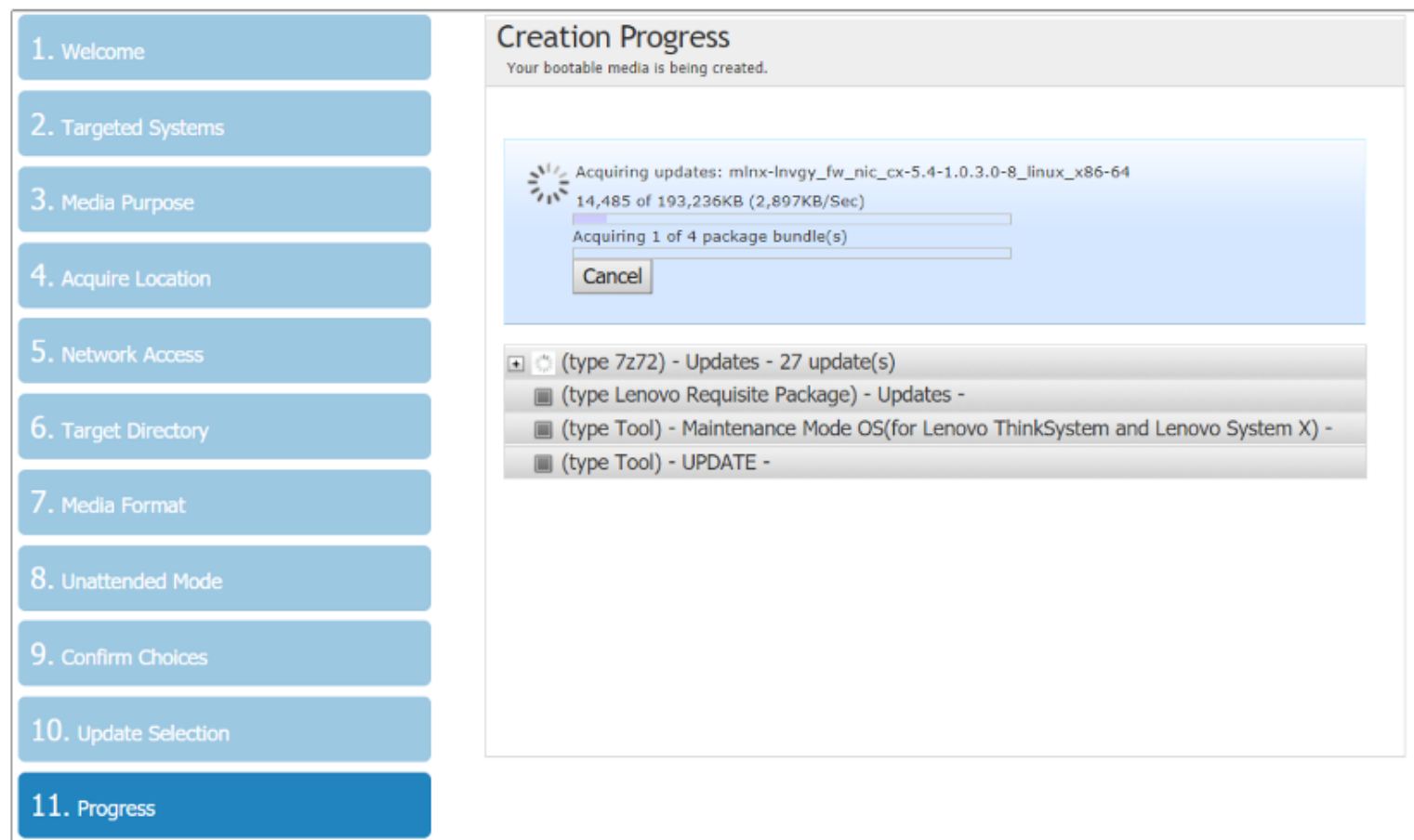
- ☒ Reboot BMC after updating BMC firmware automatically.
- ☒ Do not use unattended mode
- ☐ Use unattended mode

Step



How to create bootable media with the BoMC GUI

You can review your choices on the **9. Confirm Choices** screen and update your selection on the **10. Update Selection** screen. BoMC will start downloading the update files on the **11. Progress** screen.

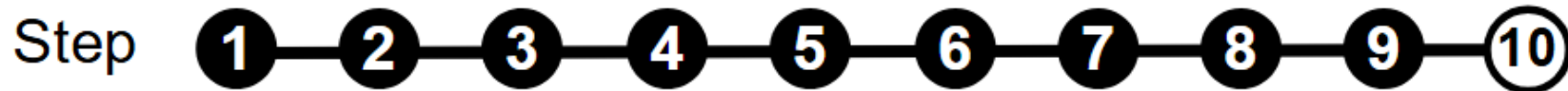
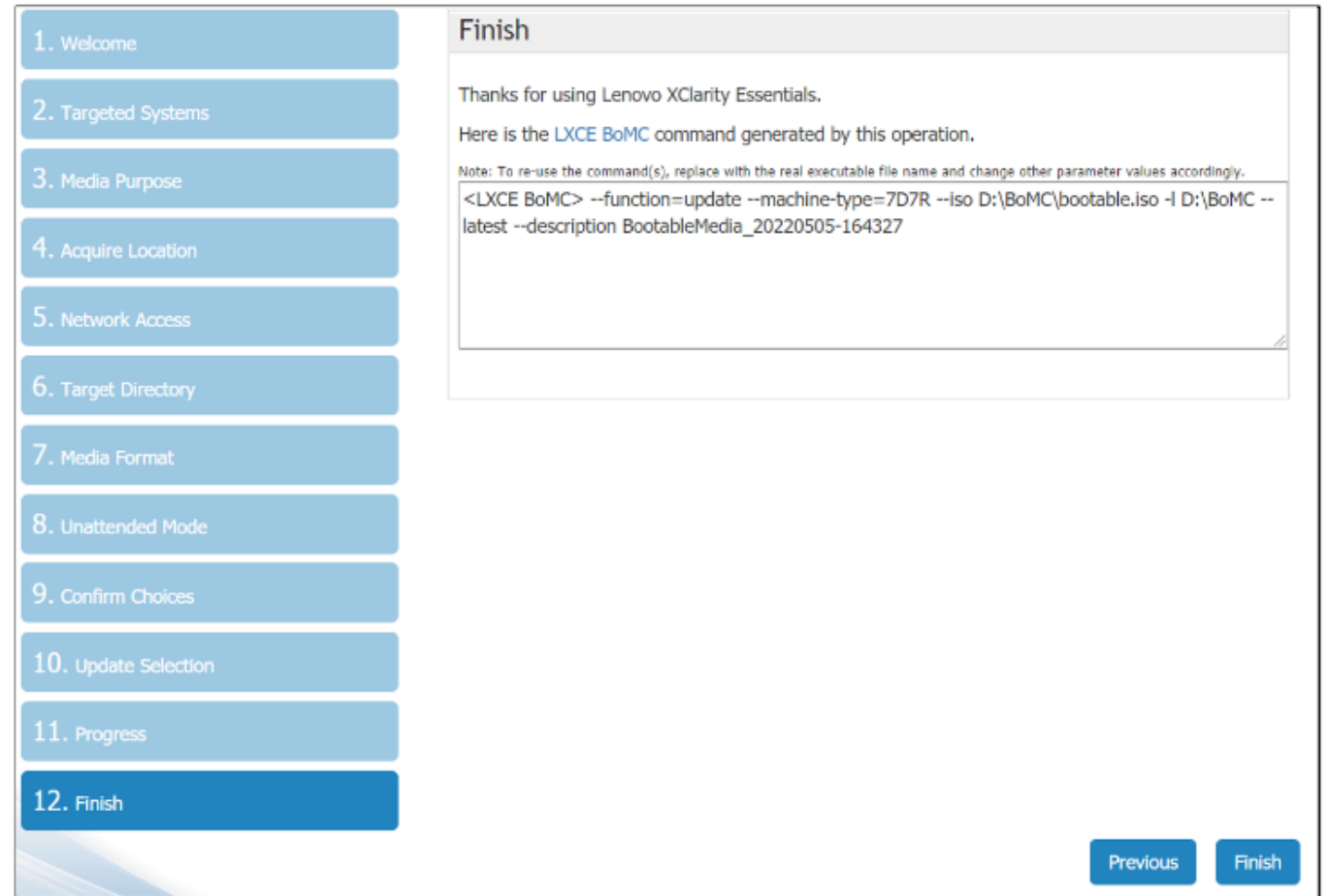


Step



How to create bootable media with the BoMC GUI

The bootable media creation process is complete. Click **Finish** to close the utility.



How to create bootable media with BoMC CLI mode

To create a bootable image to perform software updates using BoMC CLI mode, open a command prompt or terminal and navigate to the BoMC root directory.

Click the different options on the left to see the commands used to perform bootable image creation.

Check for a new
version of BoMC

Create a bootable
image using an HTTP
proxy

Create a bootable
image on a USB
device

Create a bootable
image using local files

Create a bootable
image that supports all
ThinkSystem servers

For Windows

```
lnvgy_utl_bomc_<version>_windows_i386.exe --check-update --report
```

For Linux

```
./lnvgy_utl_bomc_<version>_slesXX_i386.bin --check-update --report
```

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ThinkSystem servers

For Windows

```
lnvgy_utl_bomc_<version>_windows_i386.exe --function=update --  
machine-type=<machine type>  
-l <working directory> --iso=<output filename.iso> --proxy-  
address=<proxy ip address>  
--proxy-port=<proxy port> --proxy-user=<username> --proxy-  
password=<password>
```

For Linux

```
./lnvgy_utl_bomc_<version>_slesXX_i386.bin --function=update --  
machine-type=<machine type>  
-l <working directory> --iso=<output filename.iso> --proxy-  
address=<proxy ip address>  
--proxy-port=<proxy port> --proxy-user=<username> --proxy-  
password=<password>
```

How to create bootable media with BoMC CLI mode

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Create a bootable
image using local files

Create a bootable
image that supports all
ThinkSystem servers

For Windows

```
lnvgy_utl_bomc_11.0_windows_i386.exe --function=update --  
machine-type=<machine type>  
--usbkey=<USB drive letter> -l <working directory>
```

For Linux

```
./lnvgy_utl_bomc_<version>_slesXX_i386.bin --function=update --  
machine-type=<machine type>  
--usbkey=<USB drive letter> -l <working directory>
```

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image using an HTTP
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device

Create a bootable
image using local files

Create a bootable
image that supports all
ThinkSystem servers

For Windows

```
lnvgy_utl_bomc_11.0_windows_i386.exe --function=update --  
machine-type=<machine type>  
-l <working directory> --iso=<output filename.iso> --no-acquire
```

For Linux

```
./lnvgy_utl_bomc_<version>_slesXX_i386.bin --function=update --  
machine-type=<machine type>  
-l <working directory> --iso=<output filename.iso> --no-acquire
```


How to create bootable media with BoMC CLI mode

To create a bootable image to perform software updates using BoMC CLI mode, open a command prompt or terminal and navigate to the BoMC root directory.

Click the different options on the left to see the commands used to perform bootable image creation.

Check for a new
version of BoMC

Create a bootable
image using an HTTP
proxy

Create a bootable
image on a USB
device

Create a bootable
image using local files

Create a bootable
image that supports all
ThinkSystem servers

For Windows

```
lnvgy_utl_bomc_11.0_windows_i386.exe --function=update --  
machine-type=all-thinksystem  
-l <working directory> --iso=<output filename.iso>
```

For Linux

```
./lnvgy_utl_bomc_<version>_slesXX_i386.bin --function=update --  
machine-type=all-thinksystem  
-l <working directory> --iso=<output filename.iso>
```

BoMC full-functional ISO image

With BoMC version 12 or later, users can create a full-functional BoMC ISO image, which integrates the following functions into one ISO image:

- Firmware updates
- VPD updates
- Collecting inventory information and FFDC logs
- Configuring RAID
- Advanced system configuration
- Managing Features on Demand (FoD) keys
- Secure erase

The full-functional BoMC ISO image function applies to all ThinkSystem or System x servers which can support BoMC – it's not just for ThinkSystem V2 servers. Use BoMC v12.0.1 or later for ThinkSystem V2 servers.

Click each number in turn to see the procedure.

Step



BoMC full-functional ISO image

Download and run BoMC. Select **3. Media Purpose** and then **Full** to create the full-functional BoMC ISO image.

The screenshot shows the 'Media Purpose' step in the XClarity Essentials Bootable Media Creator. On the left is a vertical sidebar with seven steps: 1. Welcome, 2. Targeted Systems, 3. Media Purpose (highlighted in blue), 4. Acquire Location, 5. Network Access, 6. Target Directory, and 7. Media Format. The main content area is titled 'Media Purpose' and contains the following text: 'Specify what you would like your bootable media to do. Based on what you select here, you will later be prompted for the appropriate information.' There are two radio button options: 'Update' and 'Full'. The 'Full' option is selected. Below the 'Update' option is a checkbox labeled 'Enable Task AutoRun' with the description 'Allows a task to automatically begin when your bootable media is run on the target machine.' Below the 'Full' option is a checkbox labeled 'Look in a local directory for the RAID configuration file (*.ini)' with a link to 'viewSampleFile'.

1. Welcome

2. Targeted Systems

3. Media Purpose

4. Acquire Location

5. Network Access

6. Target Directory

7. Media Format

Media Purpose

Specify what you would like your bootable media to do. Based on what you select here, you will later be prompted for the appropriate information.

☐ **Update**

Will help you update the firmware on the machine types you choose. Note that this does not support IBM IntelliStation Pro, and it includes firmware updating only, and the drivers will need to be updated separately. The updating will be performed by LXCE UpdateXpress.

☐ **Enable Task AutoRun**

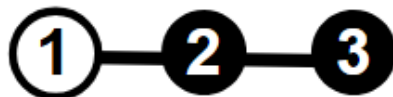
Allows a task to automatically begin when your bootable media is run on the target machine.

☒ **Full**

Will integrate all the functions in one image, including firmware update, VPD update, inventory and FFDC collection, advanced system configuration, diagnostics, RAID configuration, secure erase

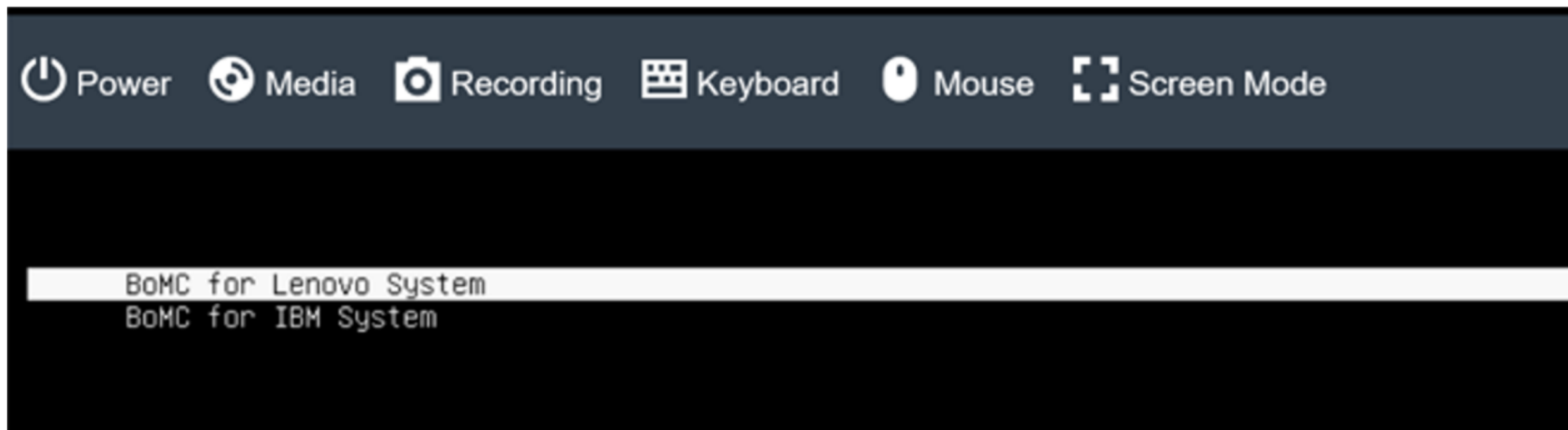
☐ Look in a local directory for the RAID configuration file (*.ini) [viewSampleFile](#)

Step

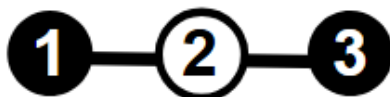


BoMC full-functional ISO image

After creating the ISO image, use it to boot the system. A BoMC for Lenovo or IBM system selection will be displayed. To continue, select the system type you are using.



Step



BoMC full-functional ISO image

A text-based user interface will be displayed. To select one of the functions, enter the corresponding number.

```
[1]      Update Firmware
[2]      Update UPD
[3]      Advanced System Configuration
[4]      Collect FFDC Logs and Inventory Information
[5]      Manage FoD Keys
[6]      RAID Configuration
[7]      Secure Erase
[8]      Enable Debug Method
[9]      Save BoMC Logs

[Q]      Quit Program and Shut Down System

* Enter your option:
Please input a valid option.
```

Step **1** — **2** — **3**

